London Ambulance Service NHS
NHS Trust

Management of Medical and Surgical Conditions – Component 5

Neurovascular Conditions

AC 2.5

Management of Neurovascular Conditions – Component 5

This session will look specific neurovascular conditions you may encounter in pre-hospital care

- Alzheimer's
- Parkinson's
- Multiple Sclerosis
- Ataxia

- Cerebrovascular Event (CVE)
- Convulsions
- Meningitis
- Coma
- Paralysis

© Department of Clinical Education & Standards

Dementia

- Dementia is a disease that affects memory, thinking and daily life
- 850,000 people in the UK have dementia, 1 in 14 over the age of 65
- The common risk groups are age, genetics or lifestyle
- The condition is often controlled by medication but no known cure

© Department of Clinical Education & Standards

Parkinson's Disease

- Parkinson's disease causes parts of the brain to become damaged over many years
- It affects approximately 1 in 500 in the UK
- Men seem slightly more likely at risk of developing the condition, but it is unclear what causes the condition to occur
- The condition is mainly treated with medication and physiotherapy

© Department of Clinical Education & Standards

Multiple Sclerosis

- Multiple Sclerosis is a condition affecting the brain and spinal cord, stopping nerve impulses travelling correctly

- It affects more than 130,000 people in the UK and more than 20,000 new cases per year are identified

© Department of Clinical Education & Standards

Multiple Sclerosis

- The condition affects mainly women and is the most common cause of disability in younger adults

- There is no cure, but it can be managed with medication

© Department of Clinical Education & Standards

Ataxia

- Ataxia is a group of disorders affecting balance, co ordination and speech
- It affects 1 in 50,000 people
- Mainly a hereditary condition, but can develop later in life
- There is no cure , but patients respond well to physiotherapy and medication

© Department of Clinical Education & Standards

Paralysis

- Paralysis is defined as partial or complete loss of muscle function
- Other terms used include –

- plegia = paralysis

-Paresis - Muscle weakness or incomplete loss of muscle function

© Department of Clinical Education & Standards

Paralysis

Terms used to identify areas affected are:

- Monoplegia – paralysis of one limb

- Hemiplegia – Paralysis of one side of body

© Department of Clinical Education & Standards

Paralysis

Terms used to identify areas affected are:

- Quadriplegia / Tetraplegia – paralysis of all four limbs
- Paraplegia – Paralysis of both lower limbs

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Cerebrovascular Event (CVE)

- Third most common cause of death in England*

  (56,000 deaths in England and Wales 11% of all deaths)
- Although most people will survive their first CVE it is the largest cause of disability
- Almost 1 million people in the UK are living with the effects of a CVE
- Half of these will depend on other people for everyday care

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

CVE risk groups

Non-Modifiable Risk Factors

- Age
- Sex
- Ethnicity
- Family history

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5



CVE risk groups



Modifiable Risk Factors

- Hypertension
- Smoking
- Atrial Fibrillation
- Diabetes
- Diet
- Physical Activity
- Alcohol Consumption
- Blood Cholesterol
- Obesity
- Previous stroke / TIA / MI





© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Stroke

- Blood supply to the brain is interrupted
- Signs and symptoms most often develop rapidly and last for at least 24 hours
- Cerebrovascular event is caused by a bleed in the brain or blockage

© Department of Clinical Education & Standards

Transient Ischaemic Attack - TIA

- Signs and symptoms suggest a Stroke, but will **resolve within 24 hours** (usually they resolve within minutes or a few hours)

- Anyone with continuing neurological signs during your management treat as a Stroke

- 20,000 TIAs occur in England every year

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Symptoms of CVE Include:

- Numbness
- Weakness or paralysis
- Slurred Speech
- Blurred vision
- Confusion / inability to communicate
- Severe headaches

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

CVE Assessment

- ABCDE for all patients
- Ambulance crews use the FAST test to diagnose stroke patients
- If a patient 'fails' 1 or more aspects of the test they are **FAST+ve** and regarded as having a stroke



© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Facial Weakness

- Ask the patient to smile or show their teeth
- Look for new lack of symmetry
- If you are unsure, ask someone who knows the patient



© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Arm Weakness

- Ask the patient to lift their arms together with their palms uppermost and hold for 5 seconds with their eyes closed
- Look for one arm drifting or falling



© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Speech Impairment

- Look for new speech disturbance
- This could be slurred speech, difficulty finding words or constructing sentences or complete inability to speak at all
- If you are unsure, ask someone who knows the patient - speech impairment can be a very subtle finding

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Time



- Why is time so important?
- With each passing minute 1.9 million neurons are destroyed
- Remember - *Although most people will survive their first stroke it is the largest cause of disability*
- Prompt treatment at a Hyper Acute Stroke Component (HASU) can save lives and reduce long-term disability

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

CVE Assessment

Your patient will be **TIME CRITICAL** if they have any –

- Major ABC problem – Nearest ED
- Positive FAST test (FAST+ve) - HASU



- In each case transport to correct receiving unit as soon as is practicable – continuing assessment en-route

© Department of Clinical Education & Standards

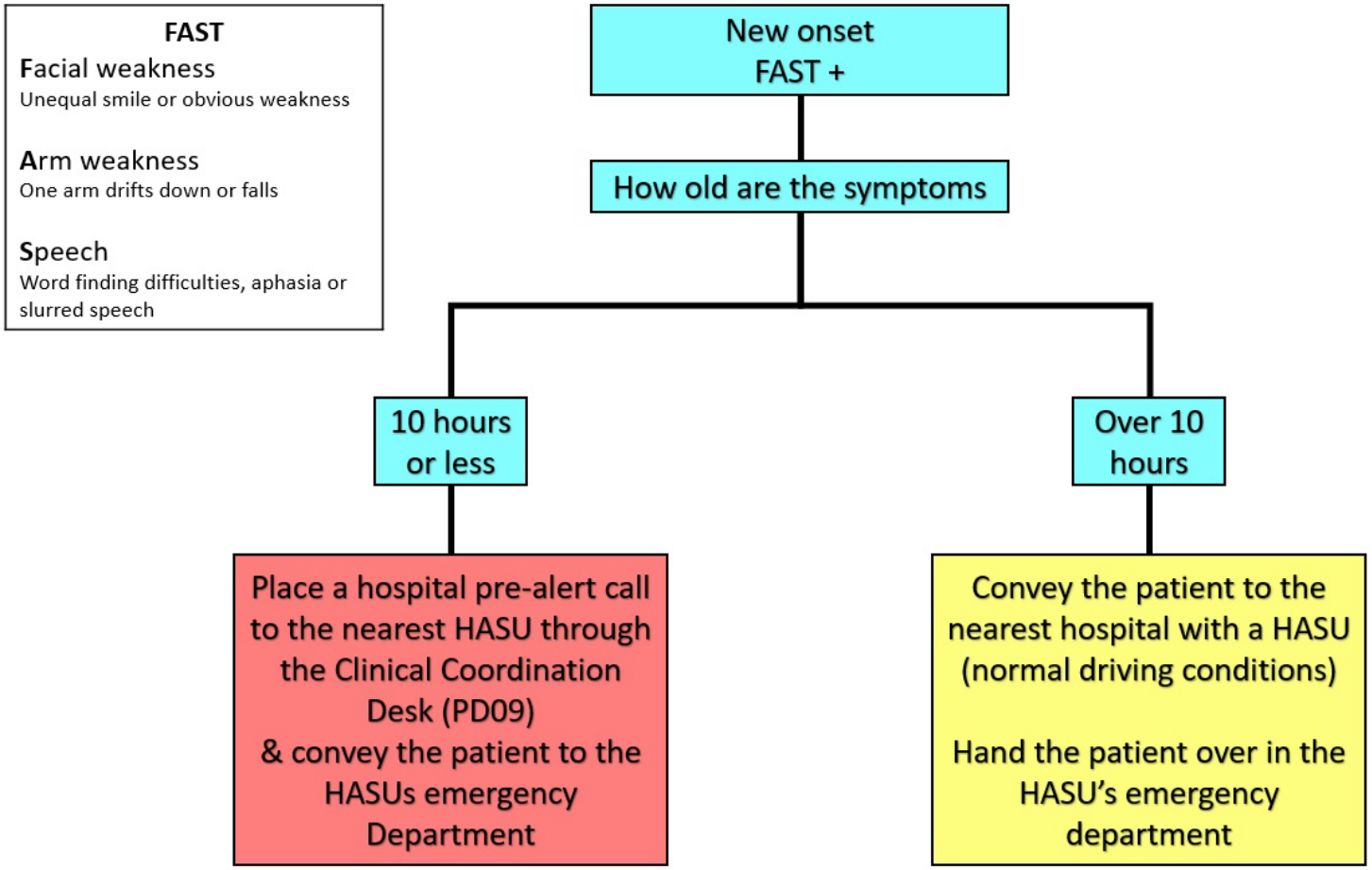
Management of Neurovascular Conditions – Component 5

CVE Management Options

- Airway protection
- Assist ventilations if RR < 10 or > 30 or inadequate
- Oxygen therapy – ONLY required if the patient is hypoxic (SPO_2 < 94%)
- Patient position – what is comfortable
- Get the patient to remain Nil By Mouth
- Reassurance
- Relatives & Friends – inform them where you will be taking the patient

© Department of Clinical Education & Standards

CVE Referral Pathways



©Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Convulsions

- Convulsion/Seizure/Fit are all words used to describe an involuntary contraction and relaxation of the muscles

- Often associated with reduced / altered level of consciousness

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Epilepsy

- Term used to describe the tendency to have recurrent unprovoked convulsions
- Still trying to be completely understood by the specialists
- Many patients are well managed by medication, while others struggle to limit the effect the condition has on their lives
- Can significantly limit cognitive development in children

© Department of Clinical Education & Standards

Types of Convulsions

Generalised

- Affect a large part of the cerebral cortex

Partial

- Affect a localised part of the cortex (although they can progress to become generalised)

© Department of Clinical Education & Standards

Generalised Convulsions

Absences

- Staring / daydreaming / absence episodes, you may notice repetitive actions such as eye blinking

- Last a few seconds with immediate recovery

© Department of Clinical Education & Standards

Generalised Convulsions

Bilateral tonic-clonic seizures - BTCS

- Tonic phase – body wide rigidity, will fall, may cry out, bite tongue as the jaw clenches.
- Clonic phase – regular jerking movements which may eventually slow / stop and the patient may be incontinent
- The patient will lose consciousness during a tonic-clonic convulsion and there is a prolonged recovery period called the Post-Ictal Phase during which the patient will be sleepy and disorientated

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Partial Convulsions

- Depending on where the convulsion is sited in the cerebral cortex -
- Patient may be conscious, have an altered level of consciousness or lose consciousness
- There may be changes in behaviour or mood
- There may be repetitive actions
- They may be localised tonic clonic activity
- And it may progress to a generalised convulsion

© Department of Clinical Education & Standards

Status Epilepticus

- Continuous convulsions lasting 5 minutes or longer

Or

- A series of convulsions (>3 or more in an hour) close together without a recovery period in between them

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Causes



Not all seizures are caused by epilepsy:

- Hypoxia
- Hypoglycaemia
- Alcohol
- Arrhythmias
- Recreational Drugs
- Eclampsia
- Kidney Failure
- Systemic Infection
- Brain Infections
- Tumour
- Stroke
- Plus more……

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Convulsion Assessment

- In 80% of cases convulsions will have stopped after 10 minutes and so may have stopped prior to your arrival
- Your assessment could therefore be of a patient who is currently having a convulsion, a patient who could be in the post-ictal phase or a patient who has recovered
- Some parts of the assessment are difficult to perform in the patient actively convulsing

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Primary Survey



Response

- Using AVPU you should be able to determine if your patient is actively convulsing, in the post-ictal phase or recovered

- If actively convulsing call for paramedic backup (if not part of crew) and start planning removal to hospital

- If the patient has a reduced LOC, protect them from injury, consider the recovery position

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Primary Survey

<u>Airway</u>

- Jaw may be clenched shut – trismus
- May have bitten tongue
- May be impossible to assess
- Do **<u>not</u>** use an OP airway adjunct in a patient who is actively convulsing

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Primary Survey



Breathing

- May be hard to assess
- Consider utilising $ETCO_2$ monitoring to assess ventilations
- Administer oxygen at 15 Lpm until a reliable SPO_2 measurement can be obtained, then titrate to 94-98%

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Primary Survey



Circulation

- May be hard to assess pulses (expect your patient to be tachycardic), but should be able to assess cap refill, skin colour and texture

- Check for blood on the floor and four more

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Primary Survey

jrcalc PLUS

Disability

- Reassess AVPU – any improvement?
- Is the convulsion being caused by a head injury?
- Senior clinician should check BM NOW!! Convulsing may be a presenting sign of hypoglycaemia

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Primary Survey



Expose / Environment

- Look for medic alert jewellery
- Check for any injuries and for incontinence
- Check for signs of underlying infection eg rash, pyrexia
- Protect your patient from injury
- Consider patient dignity

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Convulsions Management

Your patient will be **TIME CRITICAL** if they have any –

- Major ABCD problem
- Serious head injury
- Status Epilepticus
- Underlying infection



- In which case transport to hospital as soon as is practicable and continue assessment on route

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Remember

When attending seizures

"The **First**, The **Worst** or the **Longest**"

should be transferred to hospital

© Department of Clinical Education & Standards

Meningococcal Disease

- Is a general term used for bacterial infection of the meninges or infection of the bloodstream.

- Around 1200 patients are affected by meningitis per year

© Department of Clinical Education & Standards

	Septicaemia	Meningitis
 Fever and/or vomiting	✓	✓
 Severe headache		✓
 Limb/joint/muscle pain (sometimes with pain/diarrhoea)	✓	
 Cold hand and feet/shivering	✓	
 Pale or mottled skin	✓	
 Breathing fast/breathless	✓	
 Rash (anywhere on the body)	✓	✓
 Stiff neck (less common in young children)		✓
 Dislike of bright lights (less common in young children)		✓
 Very sleepy /vacant /difficult to wake	✓	✓
 Confused /delirious	✓	✓
 Seizures (fits) may also be seen		✓

Red ticks show symptoms that are more specific to meningitis and septicaemia and less common in milder illnesses

Symptoms

Meningitis and septicaemia are very dangerous and can kill in hours

They can be hard to recognise at first and symptoms can appear in any order

The first are usually
+ Fever
+ Vomiting
+ Headache
+ Feeling unwell

Limb pain and cold hands and feet often appear earlier than a rash, neck stiffness, photophobia and confusion

Not everyone gets all these symptoms

Septicaemia can occur with or without meningitis

In some cases of meningitis, a rash may not appear

If someone is ill and gets a rash, do the 'Tumbler Test'

Check for spots over the whole body

If a glass tumbler is pressed firmly against a septicaemic rash, the marks will not fade and you will see it through the glass.

If this happens get medical advice immediately

It is harder to see on dark skin, so check paler areas

Remember someone who is very ill needs medical help even if they have no rash or a rash that fades

Management of Neurovascular Conditions – Component 5

Assessment

Primary Survey

- Depending on the severity of the condition your patient may have …….
- R - a reduced level of response
- A - acute airway difficulties
- B - breathing problems – (↑RR, ↑effort, ↓SPO_2)
- C – circulation problems – (Delayed cap refill, cold skin)
- D – neurological problems – (photophobia)



© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Assessment



- <u>E – Expose</u>
- Examine for the rash –
- Haemorrhagic, non-blanching
- In an unwell patient, a non-blanching rash suggests meningococcal septicaemia
- However, do not exclude meningococcal disease in the absence of the rash, often there is NO rash

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Assessment

- Temperature
- Vomiting / Diarrhoea / Abdominal pain?
- Joint / Muscle pain



© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5



Management

Your patient will be TIME CRITICAL if they have any –

- Major ABCD problem
- Non-blanching rash or…..
- Any suspicion of meningococcal septicaemia

In which case transport to hospital as soon as is practicable and continue assessment on route

jrcalc PLUS

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Coma

- Absence of consciousness
- Loss of consciousness
- GCS of 8 and below
- It may not always be possible to find a cause for the coma and management will be supportive
- Some patients though will be unconscious from a cause that you as an ambulance crew can fix

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Causes of coma / reduced level of consciousness

- Stroke
- Convulsion
- Epilepsy
- Hypoglycaemia
- Hyperglycaemia
- Subarachnoid haemorrhage
- Overdose
- Meningitis
- Inadequate Airway
- Inadequate respiration
- Persistent hyperventilation
- Hypovolaemia
- Cardiac arrhythmias
- Clinical Shock
- Raised ICP
- Carbon monoxide
- Hypothermia

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Assessment

If any ABCD problem your patient will be TIME CRITICAL!!

- Comprehensive sets of vital signs, document GCS
- Pupils
- Thorough secondary survey to try and find a clue or a reversible cause
- 12 lead ECG
- Reassess



jrcalc PLUS

© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Assessment

jrcalc PLUS

If any ABCD problem your patient will be TIME CRITICAL!!

- Assess scene for clues, medical alerts
- Consider traumatic causes and treat appropriately
- Needle marks
- Transport to hospital as soon as is practicable and continue assessment on route



© Department of Clinical Education & Standards

Management of Neurovascular Conditions – Component 5

Any Questions?



© Department of Clinical Education & Standards